AMRITA VISHWA VIDYAPEETHAM

25AIC114

Elements of Computing – 2 (Assembly Programming)

PROJECT TITLE:

“Program Execution & Resource Monitoring System”

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Introduction

Understanding how programs execute internally is difficult for many learners.

Although concepts such as instruction execution, control flow, and memory usage are taught theoretically, the actual runtime behavior of programs is not easily visible.

This project aims to create a system that can trace program execution and monitor system resources during runtime.

Problem Statement

During program execution, it is difficult to observe:

- Execution order of instructions
- Function calls and returns
- Changes in variable values
- CPU and memory usage

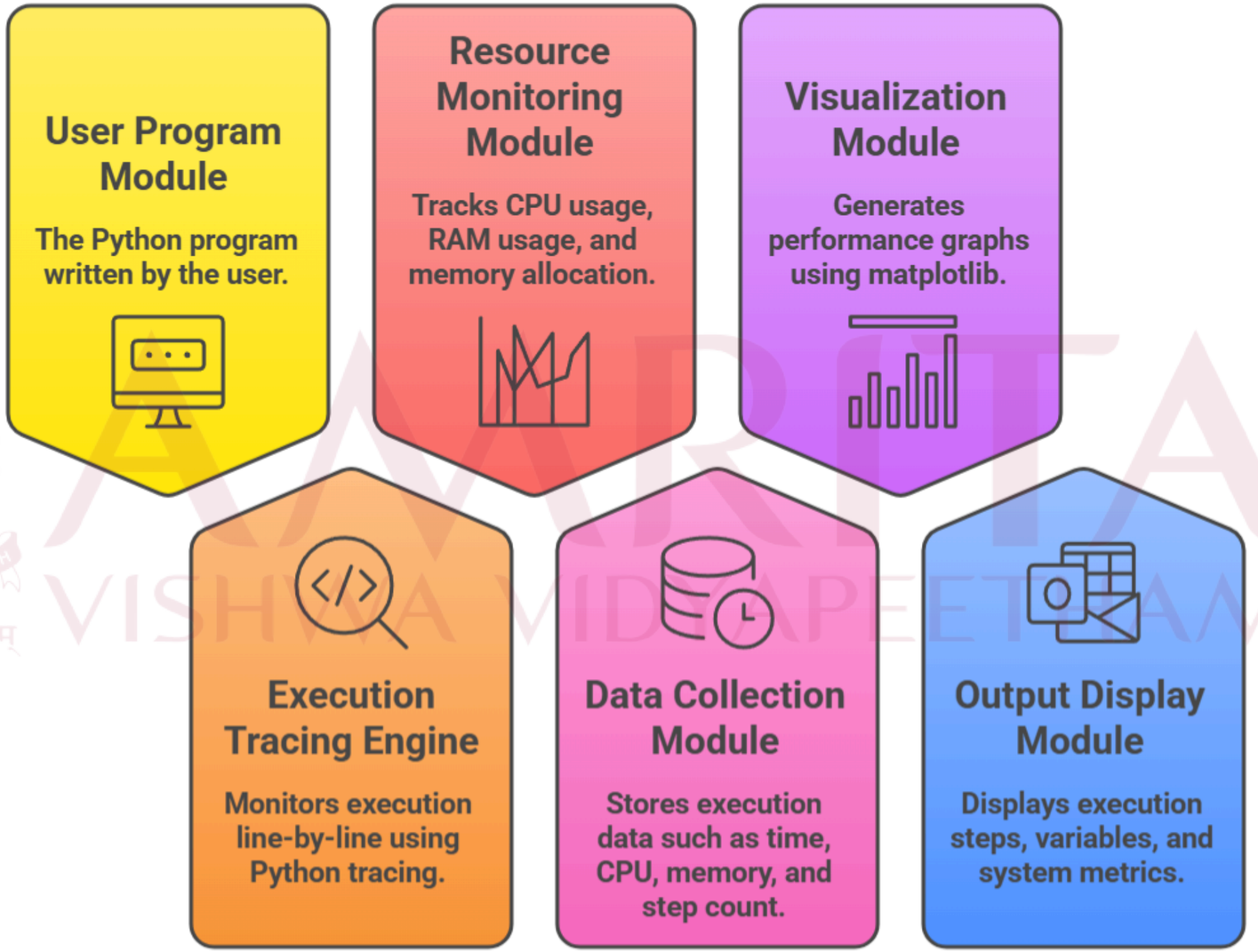
Therefore, a system is required that can monitor program execution and visualize resource usage.

Objectives

The main objectives of this project are:

- Monitor program execution step-by-step
- Track function calls and returns
- Observe variable value changes during runtime
- Measure CPU usage and RAM usage
- Measure execution time and memory consumption
- Visualize execution behavior using graphs

System Architecture



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Tools & Technologies Used

1. Programming Language
- Python
2. Libraries Used
- sys: Used for execution tracing using sys.settrace().
- os: Used for file handling and locating the user program.
- tracemalloc: Tracks memory allocation and peak memory usage.
- psutil: Used for monitoring CPU usage, RAM usage, and process memory.
- time: Used to measure execution time.
- matplotlib: Used to generate performance graphs for analysis.
3. Development Environment
- Cursor

Work Completed

The system currently performs the following functions:

Step-by-step execution tracing:

- Detection of function calls and returns
- Monitoring of runtime variable values
- Counting executed instructions (CPU steps)
- Measuring execution time
- Monitoring CPU usage
- Monitoring RAM usage
- Tracking process memory usage
- Recording peak memory usage

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System Output

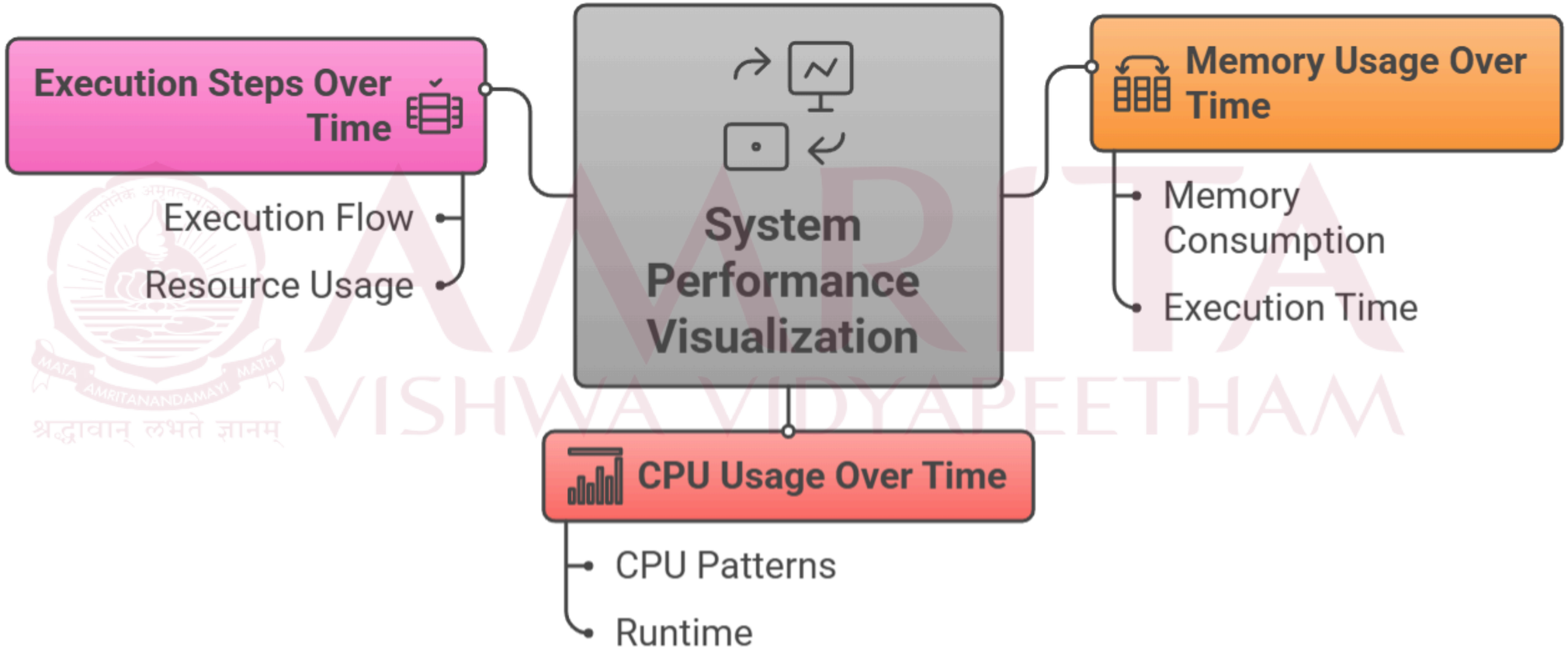
The system outputs the following information after execution:

- Total execution steps
- Function call and return traces
- Runtime variable values
- CPU usage percentage
- RAM usage percentage
- Process memory usage
- Peak memory usage
- Total execution time

This information helps analyze program behavior and system performance.

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Performance Visualization



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Future Work

Future improvements for the project include:

- Developing a graphical user interface (GUI)
- Allowing users to upload and execute programs through the interface
- Real-time execution visualization
- Program performance comparison tools
- Exporting execution reports and graphs

Individual Contributions

Member	Contribution
Poovya S	Research and literature review on program execution analysis and monitoring techniques
Abhiram	System architecture design and execution tracing implementation
Dhiren	Runtime behavior analysis, execution flow monitoring, and variable tracking
Pardhav	Resource monitoring module development (CPU, RAM, memory tracking) and code integration
Edwin	Performance graph visualization, system testing, and documentation

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Conclusion

The proposed system successfully demonstrates how programs execute internally and how system resources are utilized during execution.

By combining execution tracing with resource monitoring and visualization, the system helps users understand program behavior and performance more effectively.

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THANK YOU

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